Supplementary Material

A two-feedback Vicsek–Couzin flock: density-stratified heading correlations as the robust fingerprint of motility–noise coupling

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June 15, 2026

Order-parameter distribution at $L = 30$

The Binder dip is ambiguous on its own: a weakly first-order transition deepens U_4 through a bimodal $P(\langle\varphi\rangle)$, while a continuous transition with large γ/ν deepens it without bimodality. Only the full distribution separates the two. We histogram $\langle\varphi\rangle$ on long trajectories at $L = 30$, $N = 2000$, three seeds, 6000 measurement steps each, for the motility-only ablation at $\eta = 0.10$ and the double-adaptive model at $\eta = 0.15$ (Fig. S1a). Both histograms come out unimodal.

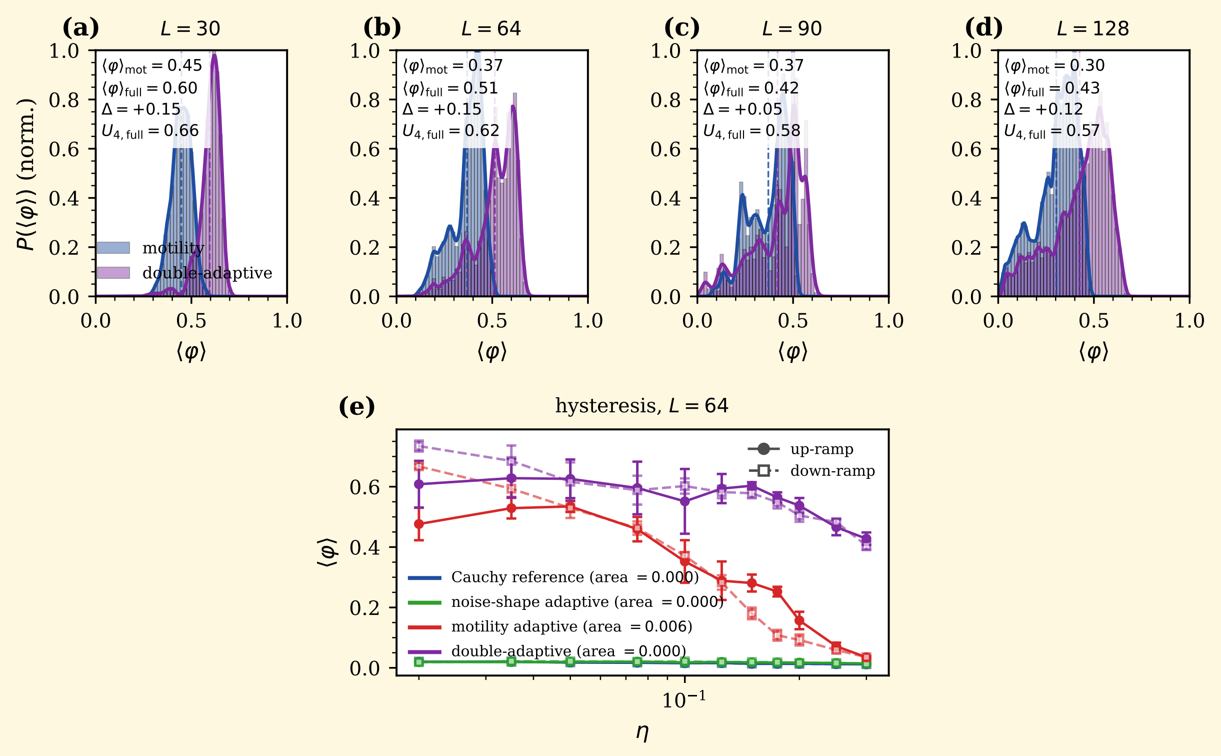

Figure S1. Order of the transition. (a)–(d) order-parameter distribution $P(\langle\varphi\rangle)$ with motility-only and double-adaptive overlaid, one panel per size (a) $L = 30$, (b) $L = 64$, (c) $L = 90$, (d) $L = 128$ (motility at $\eta = 0.10$, double-adaptive at $\eta = 0.15$), each normalised to its own peak and sharing the vertical scale; dashed lines mark the per-mode means and each panel reports the mean gap Δ and the full-mode Binder cumulant U_4. (e) quasi-static η ramp up (filled circles, solid) and down (open squares, dashed) at $L = 64$, five seeds, the two motility-active modes (hue), with the loop areas. $P(\langle\varphi\rangle)$ stays single-peaked at every size and the hysteresis loops are negligible (≤ 0.006): the deepened Binder dip is not a coexistence signature and the transition is continuous (at most very weakly first-order).

Motility-only peaks at $\langle\varphi\rangle \simeq 0.45$ with skewness -0.25, kurtosis -0.14 (a mildly left-skewed broad mode); the double-adaptive model peaks at 0.60 with a sharper core (kurtosis $+5.56$) and a long left tail (skewness -1.88). The long-trajectory Binder cumulants $U_4 = 0.65$ (motility) and 0.66 (full) sit just below the bimodal limit $U_4 \to 2/3$ and well above the Gaussian $U_4 = 0$: the trajectories are locked in a single ordered state with bounded fluctuations, not switching between two phases. The transition is at most weakly first-order here.

Rerunning the histograms at $L = 64, 90$ (five seeds, 6 000 steps) and at $L = 128$ (five seeds, 60 000 steps) in panels (b)–(d), the double-adaptive distribution sits to the right of motility-only at every size: $\Delta\langle\varphi\rangle = +0.145, +0.048, +0.121$ at $L = 64, 90, 128$ (the $L = 90$ row a seed-budget outlier already seen in s_{sep}). The full-mode U_4 stays in $[0.57, 0.62]$ and the distributions broaden into mildly platykurtic shapes at $L = 128$ ($\kappa = -0.27$ for motility, -0.20 for full) without developing two-peak structure. $P(\langle\varphi\rangle)$ remains unimodal across all four sizes.

Hysteresis across the transition

The unimodal $P(\langle\varphi\rangle)$ and the ambiguous U_4 dip leave the order of the transition unsettled; a hysteresis test settles it. We ramp η quasi-statically up and back down at $L = 64$, five seeds, carrying the configuration over from one step to the next (no re-randomisation), for the two motility-active modes (Fig. S1e). The up- and down-ramp $\varphi(\eta)$ branches overlap within error bars and the loop area is negligible: 0.006 for motility-only, 0.000 for the double-adaptive model. A first-order transition would open a finite loop; its absence, with the unimodal $P(\langle\varphi\rangle)$, places the transition in the continuous (or at most very weakly first-order) class.

Time-averaged density profile along the polar axis

The dense phase still has to be characterised geometrically: does it travel as a metric-Vicsek soliton or sit as an isotropic patch comoving with the flow? The band-soliton index b of Eq. (9) discriminates ($b \gtrsim 1.5$ for a band, $b \simeq 1$ for a patch). We measure b for all four heavy-tailed modes at their near-critical η, $L = 30$, three seeds, 4000 measurement steps (Fig. S2).

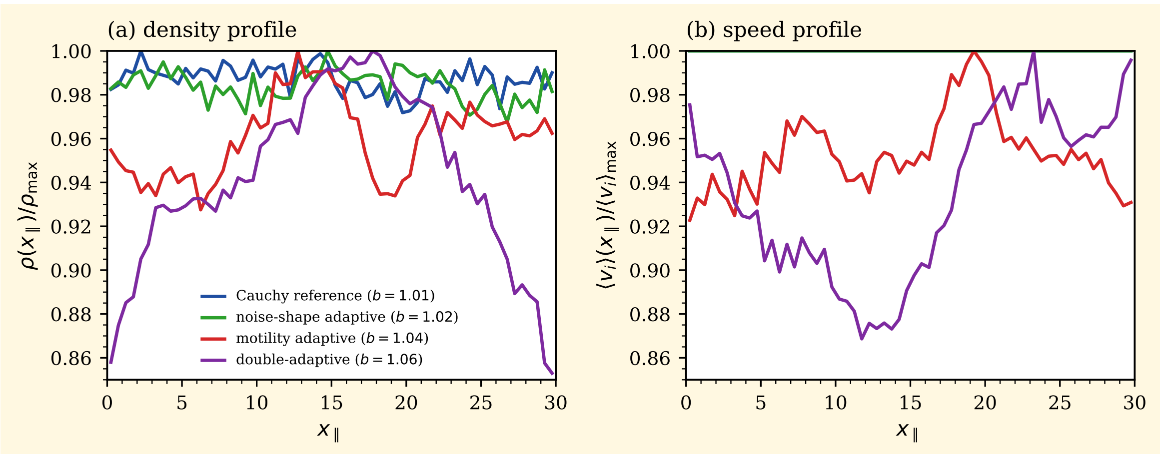

Figure S2. Time-averaged profiles along the instantaneous polar-flow axis $x_\parallel$ at $L = 30$, three seeds, 4000 measurement steps, the four modes overlaid at their respective near-critical η. (a) density normalised to its peak, with the band-soliton index b per mode in the legend. (b) local speed normalised to its peak. Every profile is flat ($b \simeq 1$), so the motility-built dense phase is an isotropic comoving patch, not a travelling Vicsek band.

All four profiles are flat to within 7 % ($b = 1.01, 1.02, 1.04, 1.06$ in the order Cauchy, noise-shape, motility, full). None supports a metric-Vicsek band: the motility-built dense phase either reorganises faster than its polar axis or is isotropic in the comoving frame. The full mode is marginally the most cohesive, consistent with the Gaussian rectification tightening the patch from within, but the increment is well below the soliton threshold.

Why $\chi_{\max}(L)$ is a crossover, not an exponent

The log–log slopes $a_{\text{motility}} = +1.29$ and $a_{\text{full}} = +1.95$ of Table 1 are steep enough to invite a critical reading, so we test that reading directly on the homogeneous ten-seed series (Fig. S3). A genuine power law would hold a *constant* local slope $a_{\text{eff}} = d\ln\chi_{\max}/d\ln L$ across the range; instead, for both motility-active modes a_{eff} overshoots its own global fit at intermediate sizes and then falls back toward zero in the largest interval ($L = 90 \to 128$ gives $a_{\text{eff}} = +0.07$ for the full mode and $+0.13$ for motility-only), while the two fixed-motility references stay pinned at $a_{\text{eff}} \simeq 0$ (panel a). The growth is therefore a finite-size transient that is already saturating at the top of our window, not a clean asymptotic scaling.

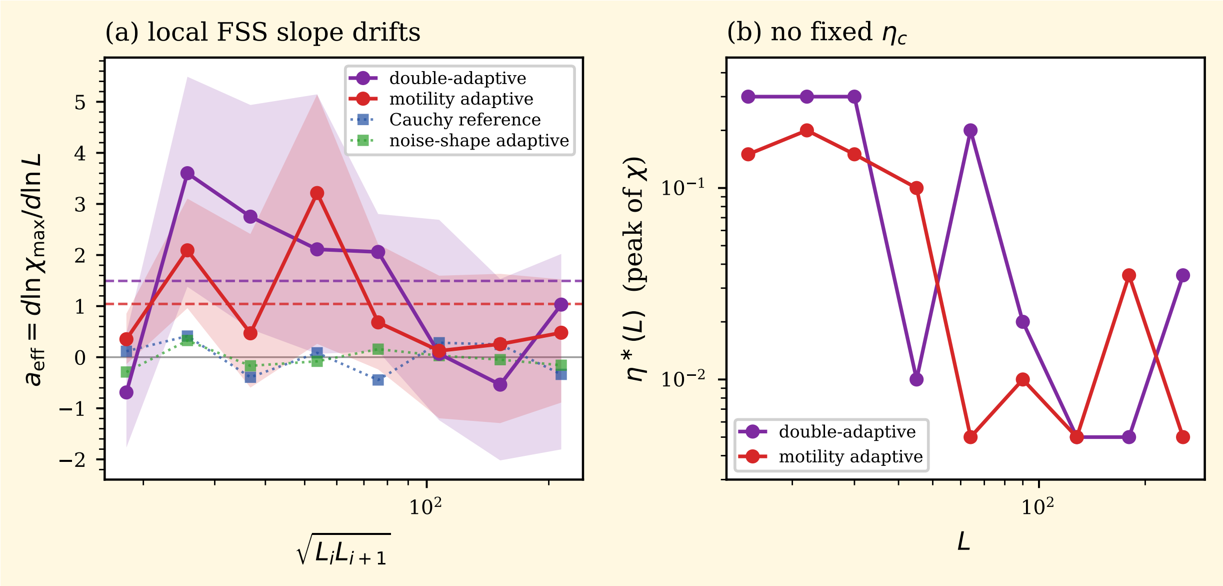

Figure S3. Finite-size-scaling diagnostic on the homogeneous ten-seed series. (a) local effective slope $a_{\text{eff}} = d\ln\chi_{\max}/d\ln L$ between consecutive sizes, plotted at the geometric-mean size; dashed lines mark each mode's global log–log fit. The two motility-active modes overshoot their global slope at intermediate L and collapse toward zero in the largest interval, whereas the Cauchy and noise-shape references hug $a_{\text{eff}} \simeq 0$ throughout; bands are 95% seed bootstraps ($B = 4000$). (b) position $\eta^*(L)$ of the susceptibility peak: instead of converging to a size-independent η_c, it migrates toward the small-η boundary, so the scaling variable $(\eta - \eta_c)L^{1/\nu}$ of a standard collapse cannot be defined here.

A standard collapse χ/L^a against $(\eta - \eta_c)L^{1/\nu}$ needs a size-independent critical point η_c. Panel (b) shows there is none: the susceptibility peak $\eta^*(L)$ does not settle but drifts toward the smallest accessible noise as L grows, the signature of an order-from-density (MIPS-like) growth in which χ keeps rising as $\eta \to 0$ rather than diverging at a finite η_c. With the slope still drifting and the peak still migrating at $L = 128$, we report a_{full} and a_{motility} as the growth rate of a finite-size phase-separation susceptibility, and read $\Delta_7 = +0.63$ as a super-additive crossover excess rather than a difference of asymptotic exponents. Controlled runs at $L = 180$ and 256 would test whether the excess survives once the transient has fully saturated.